

04

LECTURE

HEALTH CARE

MT 1, Second Semester LECTURE 04 - Ms. Bianca Marie E. Racal, RMT



COMMUNICABLE DISEASE CONTROL

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LEARNING OBJECTIVES

- Define and differentiate communicable versus non-communicable diseases
- Define and differentiate acute versus chronic disease
- To understand the communicable disease model, chain of infection, and modes of transmission, along with definition of some significant terms
- Prevention, intervention, control, and eradication of disease

I. COMMUNICABLE VS. NON-COMMUNICABLE DISEASE

Communicable Diseases or Infectious Diseases

- Diseases that are caused by biological agents that are transmitted from **transmissible** from one person to another
- And infection begins when an agent enters, grows, and reproduces within the host's body
- Can be transmitted through direct contact, indirect contact, airborne transmission, or vectors

Noncommunicable Diseases or Non-Infectious Diseases

- Diseases that **cannot be transmitted**
- Often called **multi-causal diseases**, which can be caused by:
 - Genetic
 - Environmental
 - Lifestyle

II. ACUTE VS. CHRONIC DISEASES

Acute Diseases | Short-term

- Peak severity of symptoms occurs
- Subsides within three months or less
- Recovery is usually complete
- Example: **Chickenpox**

- You can only be infected once since you develop immunity after infection

Chronic Diseases | Long-term

- Symptoms continue for longer than three months, often lasting for years or a lifetime
- Recovery is slow and sometimes incomplete
- Can be managed and not always cured
- Example: **Diabetes**

CLASSIFICATION OF DISEASES

Type of Disease	Examples
<i>Acute Diseases</i>	
Communicable	Common cold, pneumonia, mumps, measles, pertussis, typhoid fever, cholera
Noncommunicable	Appendicitis, poisoning, injury (due to motor vehicle crash, fire, gunshot, etc.)
<i>Chronic Diseases</i>	
Communicable	AIDS, Lyme Disease, tuberculosis, syphilis, rheumatic fever following streptococcal infections, hepatitis B
Noncommunicable	Diabetes, coronary heart disease, osteoarthritis, liver cirrhosis due to alcoholism

Table 1. Classification of Disease

III. COMMUNICABLE DISEASE

A. DEFINITION OF TERMS

Infectivity

- Ability of a biological agent to enter, grow, and multiply in a host and spread to other hosts
- How and how fast does it spread?

Pathogenicity

- Infectious agent's ability to produce a disease
- How harmful is the infectious agent?
- Example: **Omicron variant of COVID-19** has a high infectivity but low pathogenicity

B. COMMUNICABLE DISEASE MODEL

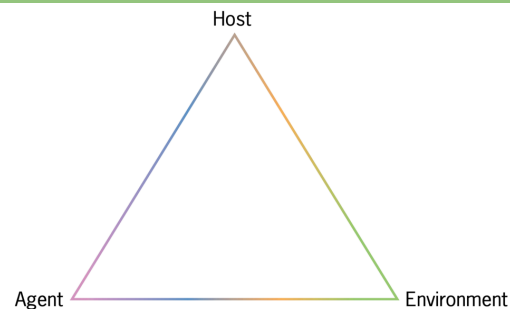


Figure 1. Communicable Disease Model

Elements of the Communicable Disease Model

- **Agent**
 - Cause of a disease
 - Element present for disease to occur
 - Example: **Influenza Virus**

- **Host**
 - Any susceptible organism invaded by the infectious agent
 - Example: **Humans, Animals, or Plants**
- **Environment**
 - Includes all other factors that help regulate disease transmission
 - Can be:
 - Physical Factors
 - Biological Factors
 - Social Factors

Biological Agents of Disease

TABLE 4.3 Biological Agents of Disease		
Type of Agent	Name of Agent	Disease
Viruses	Varicella virus	Chickenpox
	Human immunodeficiency virus (HIV)	Acquired immune deficiency syndrome (AIDS)
	Rubella virus	German measles
Rickettsiae	<i>Rickettsia rickettsii</i>	Rocky Mountain spotted fever
	<i>Vibrio cholerae</i>	Cholera
	<i>Clostridium tetani</i>	Tetanus
Bacteria	<i>Yersinia pestis</i>	Plague
	<i>Borrelia burgdorferi</i>	Lyme disease
	<i>Entamoeba histolytica</i>	Amebic dysentery
Protozoa	<i>Plasmodium falciparum</i>	Malaria
	<i>Trypanosoma gambiense</i>	African sleeping sickness
	<i>Tinea cruris</i>	Jock itch
Fungi and yeasts	<i>Tinea pedis</i>	Athlete's foot
	<i>Wuchereria bancrofti</i>	Filariasis (elephantiasis)
	<i>Onchocerca volvulus</i>	Onchocerciasis (river blindness)
Nematoda (worms)		

Figure 2. Biological Agents of Disease

IV. CHAIN OF INFECTION

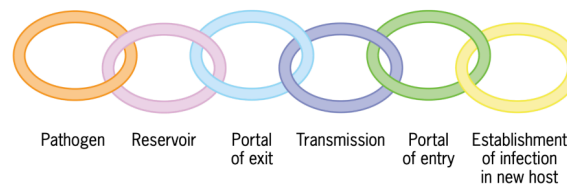


Figure 3. Chain of Infection

- Process by which an infectious disease spreads from one host to another through a series of steps

- Removing any of the links in the chain breaks disease prevention
 - Best way to break the chain: **Hand Hygiene**
- There are six elements in the chain of infection, which have their own roles and are in order
- The variations of the chain of infection depend on the:
 - Disease agent
 - Environmental Condition
 - Infectivity
 - Host susceptibility

What Happens in a Chain of Infection?

- The pathogenic disease-producing agent leave its reservoir or infected host, via the portal of exit
- Transmission occurs by direct or indirect manner
- The pathogenic agent enters the susceptible host through a portal of entry to establish infection

A. ELEMENTS OF THE CHAIN OF INFECTION

Infectious Agent

- The microorganism that causes disease
 - Bacteria, virus, fungus, or parasite
- Example: **Influenza virus** causes the **flu**

Reservoir

- The natural habitat where the infectious agent lives, grows, and multiplies
- Can be humans, animals, soil, or contaminated surfaces
- Example: **Mosquitoes** serve as reservoir for **malaria**

Portal of Exit

- The route through which the infectious agent leaves the reservoir
- Can be through:
 - Respiratory secretions
 - Coughing and Sneezing
 - Blood
 - Feces
 - Open Wounds
- Example: **Tuberculosis** exits through **respiratory droplets** when an infected person coughs

Mode of Transmission

- The way the infectious agent spreads from one host to another
- Can be:
 - Direct
 - Person-to-Person Contact
 - Indirect
 - Contaminated Surfaces, Airborne, Vector-borne
 - There are a lot of mode of indirect transmissions
- Example: **COVID-19** spreads through **respiratory droplets** and **contaminated surfaces**

Portal of Entry

- The route through which the infectious agent enters a new host
- Common portals include:
 - Respiratory Tract
 - Gastrointestinal Tract
 - Mucous Membranes
 - Broken Skin
- Example: **HIV** enters through **blood contact** or **sexual transmission**

Susceptible Host

- An individual who lacks immunity or resistance to the infectious agent
- Factors affecting susceptibility include:
 - Age
 - Underlying Health Conditions
 - Weakened Immune Systems
- Example: **Elderly individuals with weak immune systems** are more susceptible to **pneumonia**

B. DEFINITION OF TERMS

Case

- Person who is sick with a disease

Carrier

- A person or animals that harbors a specific communicable agent in the absence of discernible clinical disease
- More dangerous than a case because they may **show no symptoms** or are **asymptomatic**
- Serves as a potential source of infection to others

Zoonosis

- Communicable disease transmissible under natural conditions **from vertebrae animals to humans**
- Example: **Rabies**

Anthroponosis

- Diseases that **only infect humans**

C. PORTALS OF EXIT FOR SOME DISEASES

Portal of Exit	Disease	
Respiratory Tract	- Common Colds - Influenza	- Measles - Tuberculosis
Urogenital Tract <i>STIs</i>	- Gonorrhea - Syphilis	- Herpes - AIDS
Digestive Tract	- Amebic Dysentery - Shigellosis	- Polio - Typhoid Fever - Cholera
Skin	- Ringworm	- Jock Itch
Transplacental	- Rubella Virus - Syphilis	- Spirochetes - Hepatitis B Virus

Table 2. Portals of Exit for Some Diseases

D. MODES OF TRANSMISSION

Direct Transmission

- Direct transfer of the disease agent between the infected and susceptible individual
- **Examples:**
 - Touching
 - Biting
 - Kissing
 - Sexual Intercourse
 - Direct Projection (Droplet Spread)
- **Diseases:**
 - Sexually Transmitted Infections (STIs)
 - Rabies

Indirect Transmission

- Airborne
 - Dissemination of microbial aerosols to suitable portal of entry
 - **Diseases:**
 - Tuberculosis
 - Influenza
 - Measles

TYPES OF INDIRECT TRANSMISSION

Vehicle-Borne Transmission

- **Contaminated materials or objects** (fomites) serving as vehicles where communicable agents are transferred to a susceptible host.
- **Examples:** toys, handkerchiefs, utensils, water, food, milk, blood, serum, plasma, organs, tissue

Vector-Borne Transmission

- Transfer of a disease by a **living organism** such as mosquitoes, flies, and ticks.
- Transmission may be **mechanical or biological**
- **Mechanical Transmission**
 - Contaminated mouth parts or feet of vector
- **Biological Transmission**
 - Multiplication or developmental changes of agent in the vector before transmission
 - Example: **Mosquitoes**
 - Transmit viruses, causing
 - Yellow Fever
 - Dengue Fever (most common in the Philippines)
 - West Nile Virus
 - Zika Virus

Example: Ticks

- Can cause
 - Rocky Mountain Spotted Fever
 - Relapsing Fever
 - Lyme Disease (most common disease spread by ticks)

V. PREVENTION, INTERVENTION, CONTROL, AND ERADICATION OF DISEASES

Prevention

- Implies planning for and taking action to prevent or forestall the occurrence of an undesirable event
- Prevention is better than cure
- Example: **Immunization**

Intervention

- Taking of action **during an event**
- Example: **Taking an antibiotic** to cure an infection

Control

- **Containment of a disease**
- May include prevention and intervention measures
- Limiting of transmission of a communicable disease

Eradiation

- **Uprooting or total elimination** of a disease from the human population
- An elusive goal, one that is rarely achieved in public health
 - Smallpox is the only communicable disease that has been eradicated

BOX 4.4 Community Health in Your World: Smallpox Eradication

October 2015 marked **38 years since the last naturally acquired case of smallpox** in the world. This last case occurred in Somalia in October 1977. Although two cases of smallpox were reported in the United Kingdom in 1978, these were associated with a research laboratory and did not represent a natural recurrence.

Smallpox is caused by the **varicella virus**. In its severest form, it is a **disfiguring and deadly** disease. Manifestations of the disease include **fever, headache, malaise, and prostration**. A **rash** appears and covers the body, and there is **bleeding** into the skin, mucous linings, and genital tract.

The circulatory system is also severely affected. Between **15% and 40% of cases die**, usually within **2 weeks**. Survivors are **terribly scarred for life and are sometimes blinded**.

Mass vaccinations and case-finding measures by the World Health Organization (WHO), with financial support from the United States, led to the eradication of smallpox from the world. Why was it possible to eradicate smallpox? Why have we been unable to eradicate any other diseases since 1977? Do you think we will ever be able to do so? If so, what disease will be eliminated next?

Figure 4. Smallpox Eradication

A. LEVELS OF PREVENTION

Primary Prevention

- Break the chain of infection or interrupt the disease transmission cycle
- Forestall the onset of illness/injury during the Prepathogenesis Period (before the disease happens)
- Prevention is better than cure
- Examples:
 - Health Education and Health Promotion Programs
 - Immunization
 - Chlorination of Water Supply

Secondary Prevention

- Preventive measures lead to an early diagnosis
- **Health Screenings**
 - Detect disease in early pathogenesis stage, permitting intervention (through treatment) and limiting disability
- Purpose is not to diagnose disease

Tertiary Prevention

- Re-train, re-educate, and rehabilitate the patient who already incurred a disability
- Focuses on more in **rehabilitation**
 - So that disease will not happen again

- Therapy for patient who suffered from myocardial infarction or cerebrovascular spasm

VI. THE NATURAL HISTORY OF ANY DISEASE OF HUMANS

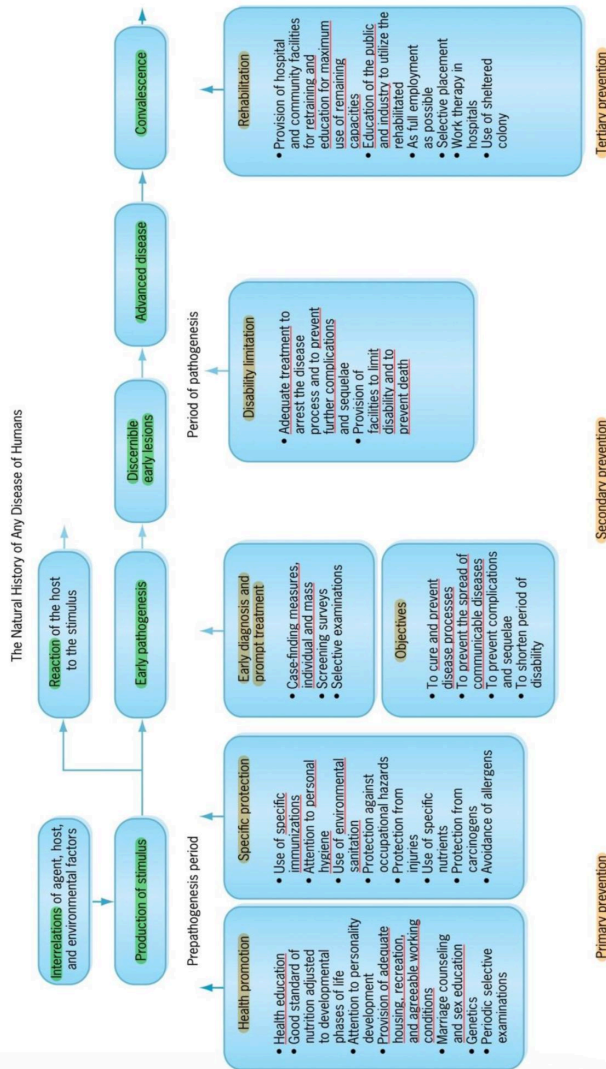


Figure 5. Natural History of Any Disease of Humans

Figure 5 Breakdown

- Prepathogenesis Period - **Primary Prevention**
 - **Health Promotion**
 - Health Education
 - Sex Education
 - Marriage Counselling
 - **Specific Protection**
 - Immunization
 - Personal Hygiene
 - Environmental Sanitation
- Early Pathogenesis - **Secondary Prevention**
 - **Early Diagnosis and Prompt Treatment**
 - Screening Surveys
 - Selective Examinations
 - **Disability Limitation**
 - Adequate Treatment
 - Provision of Health Facilities for Treatment
- Convalescence - **Tertiary Prevention**
 - **Rehabilitation**
 - Selective Placement
 - Work Therapy

VII. APPLICATION OF THE LEVELS OF PREVENTION TO COMMUNICABLE DISEASES

A. PRIMARY PREVENTION

Ultimate Goal

- Break the chain of infection or interrupt the disease transmission cycle

Measures that Can Be Done By:

- Government
 - Water Supply Chlorination
 - Immunization Programs
 - Proper Solid Waste Disposal

- Citizens
 - Handwashing
 - Proper Cooking of Food
 - Adequate Clothing and Housing
 - Use of Condoms
 - Getting Immunized

2 Types of Immunity

• Active Immunity

- When exposure to disease-causing organisms prompts the immune system to develop antibodies against the disease
 - Only happens when there is disease already
- Example: **Vaccination**
 - Vaccines contain weakened, inactivated, or harmless parts of a pathogen that train the immune system to recognize it, so the body can respond quickly and effectively if exposed to the real infection.

• Passive Immunity

- When a person receives antibodies from their mother

B. SECONDARY PREVENTION

Involves:

- Self-diagnosis and self-treatment with non-prescription medications or home remedies
- Diagnosis and treatment with an antibiotic prescribed by a physician

Usually Aimed at:

- Controlling or limiting the extent of an epidemic
 - Doing something to limit disability
- Maintaining records and complying with regulations requiring reporting of notifiable diseases and investigating cases and contacts

Control Measures

- **Isolation**
 - The separation for the period of communicability of infected persons from those who are susceptible
 - To prevent direct or indirect transmission of communicable agents
- **Quarantine**
 - Limitation of freedom of movement of those who have been exposed to disease and may be incubating it
- **Disinfection**
 - The killing of communicable disease agents outside the host

C. TERTIARY PREVENTION

Preventive Measures

- Convalescence from Infection
- Recovery to Full Health
- Return to Normal Activity

Aimed at:

- Preventing the recurrence of an epidemic
- Example: Proper removal, embalming, and burial of the dead infected with a communicable disease

VIII. GREAT PUBLIC HEALTH ACHIEVEMENTS: 2001–2010 IN THE CONTROL OF INFECTIOUS DISEASE

Vaccine

- Has a significant impact on the burden of disease
- Reduces significant hospitalizations and death

Improvements in Public Health Infrastructure, Innovative and Targeted Prevention Efforts, and Advances in Laboratory Techniques and Technology

- Enhanced Disease Surveillance
- Expanding the age range for HIV Screening recommendations
- 2004: The CDC has officially declared the USA free from canine rabies

Public Health Preparedness and Response

- Planning is the best way to minimize the potential impact of a public health threat

IX. REFERENCES

- McKenzie, J., Pinger, R., and Seabert, D. (2018). **An Introduction to Community and Public Health**. 9th Edition